

Research Project - Workday #1

General Project Reminders

- You will be giving a 5-7 minute presentation the week before Spring Break on your topic.
- Your project should be built around a diagram, graph, model, etc. that substantially contributes to our understanding of the topic. A presentation aid (like PowerPoint, Prezi, etc.) is not a visual!
- In addition to your visual, you will likely want to have a presentation aid like a PowerPoint, a poster, or something similar.

What to do before we go to the library

Step 1

Start by writing down your topic from <https://go.rohrbachscience.com/project-topics>. Make sure to include any information in parentheses, as that information may help guide your research.

Step 2

Please download and edit the template posted at <https://go.rohrbachscience.com/bib-template>.

Step 3

Open up Mrs. Kovach's Research Pathfinder at <https://go.rohrbachscience.com/pathfinder>. There are excellent links to help you get started on your research.

Your Assignment: An Annotated Bibliography

Your goal for today is to find some good sources about your topic and to start learning about it. The evidence that you've met the goal will be your Annotated Bibliography.

You need to find **three sources** about your topic. Each source needs:

1. An APA-style citation
2. A paragraph explaining what you learned from that source
3. One sentence evaluating the source.
4. One sentence explaining how your source will help you explain your project topic.

Please see the last few pages of the Research Pathfinder includes instructions on how to cite your three sources and complete their annotations.

Your Annotated Bibliography should be double spaced and have a hanging indent. It should be uploaded to Schoology by **Sun, Feb 8 at 11:59pm**.

Name:

Date:

Period:

Questions to Guide Your Research

These are questions to help you if you get stuck. I do not need you to turn in answers to these questions.

1. What will my visual be? A diagram? A model? A graph? A flow chart?
2. What is the science behind my topic? Why is it interesting/important?
3. (*if you have a scientist*): What scientific theories is my person responsible for?
4. (*if you have a building project*): How will I build it? What materials will I need?

Project Timeline

Sun, Feb 8 (11:59pm on Schoology)

Annotated Bibliography due

Fri, Feb 20 (Pd 2, 3, 5) / Mon, Feb 23 (Pd 1, 4, 6)

Library Research Day #2

Week of Mar 16-20

Presentations in class

Sat, Mar 21

Spring Break!

Types of Citations

Please make sure to set up a hanging indent on your document. To create a hanging indent in Microsoft Word, highlight your text, right-click, and select **Paragraph**. Under the “Indentation” section, go to **Special** and select **Hanging** from the dropdown menu, setting it to 0.5 inches. If you use the template provided, the hanging indent is already set up for you.

Professional Website with an author:

Last Name of Author, First initial*. (Year, Month Date[†]). *Title of Page*. Site Name. Website address

Example:

Hubilla, C. (2023, July 27). *Solving the mystery of Prince Rupert’s Drop: Scientists reveal the secret behind its incredible strength*. Science Times. <https://www.sciencetimes.com/articles/44555/20230627/solving-mystery-prince-rupert-drop-scientists-reveal-secret-behind-incredible.htm>

*If there are two authors, list by their last names and initials. Separate author names with a comma. Use the ampersand (&) instead of “and.” *Example: “Soto, C. J., & John, O. P.”*

For 3 to 20 Authors, list by last names and initials; commas separate author names, while the last author’s name is preceded again by ampersand. *Example: “Nguyen, T., Carnevale, J. J., Scholer, A. A., Miele, D. B., & Fujita, K.”*

Professional Website with no author:

Sponsor of Website. (Year, Month Date[†]). *Title of Page*. Website address

Example:

Physics Classroom. (n.d.). *Rainbow Formation*. <https://www.physicsclassroom.com/class/refrn/Lesson-4/Rainbow-Formation>

[†]If there is not a publication date, you may write “(n.d.)” (for “no date”).

Database Citation:

The citation is located at the end of the article. (Make sure to select “APA style.”)

Example:

Toupin, L. (2014). Rainbows. In K. L. Lerner & B. W. Lerner (Eds.), *The Gale Encyclopedia of Science* (5th ed.). Gale. <https://link.gale.com/apps/doc/CV2644031869/SCIC?u=avon12&sid=bookmark-SCIC&xid=2ce15470>

Annotated Bibliography

Pieces to include:

1. APA Citation.
2. A summary about the source.
3. A sentence that evaluates the source.
 - How do you know it is a reliable source?
 - Is it a professional website or a database article?
4. A sentence that explains why this source is helpful to your research.

Example:

Andrews, T. M. (2017, May 17). *The head of this teardrop-shaped glass can withstand bullets.*

Washington Post. <https://link.gale.com/apps/doc/A491862572/SCIC?u=avon12&sid=bookmark-SCIC&xid=d68948b4>

This resource has information about Prince Rupert's drops. These drops resemble tadpoles. The head of the drop is extraordinarily strong and can withstand being hit by a hammer or a shot from a .22-caliber handgun. However, the drop's tail is very fragile and once it is broken the whole drop will shatter into fine dust. Although the drops have been around since the 17 Century, it was only recently that researchers from Purdue University, University of Cambridge, and Tallinn University of Technology were able to measure the compressive strength of the drops by using a technique called integrated photoelasticity. This is a reliable source of information, because it is a database article from the Gale in Context: Science database and was originally published in the Washington Post. This source is extremely helpful to use for research purposes because it has necessary Information about compressive forces and tensile forces, and how they apply to the stress distribution within Prince Rupert's drops.